МЕЖДУНАРОДНЫЙ УНИВЕРСИТЕТ INTERNATIONAL UNIVERSITY
ХАЛЫҚАРАЛЫҚ УНИВЕРСИТЕТІ

Faculty of Computer Technology and Cyber Security

Laboratory work №8
By discipline: “Fundamentals of Logic Design”
On the topic: “The Master Sword”

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Introduction

The topic of the last laboratory work was to choose one item among popular games. As a person who has never played video games and does not know about weapons, I chose a sword. Master sword from “The Legend of Zelda”.

Workflow

There are two parts to the construction and creation of a sword, blade and hilt.



Figure 1.1 – The Master Sword

The first part of the work is creating the blade. I create two additional planes to the left and right of the Front plane at a distance of 1.50 mm. On each plane, a sketch of a sword is created around the perimeter. Three sketches are combined due to the loft boss/base feature.

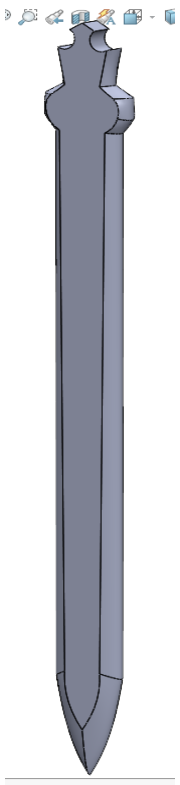


Figure 1.2 – The blade

To create the cutout on the blade, draw three triangles using centerline. The side length of each triangle is 2.20 mm and the distance between them is 0.10 mm. Using the extruded cut tool, a cutout is created on one side.

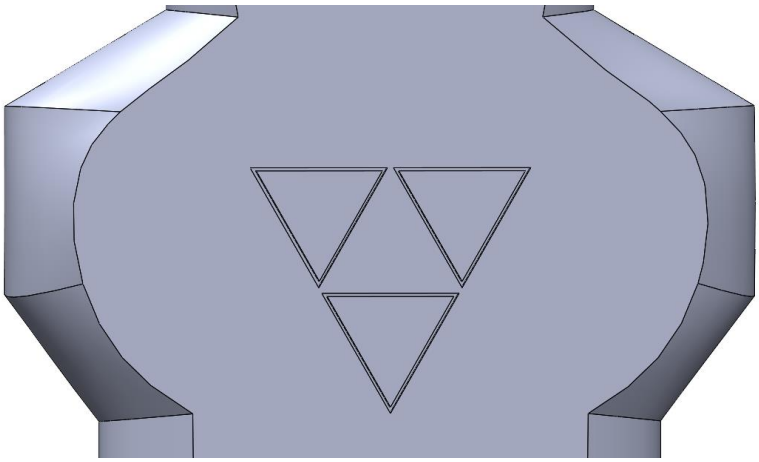


Figure 1.3 – Triangles

I drew two balls at the beginning of the guard with a radius of 1.5mm. In the main part, I used the extruded boss/base function on a two-dimensional drawing. In addition, for a neater transition from blade to guard, I rounded off the edge.

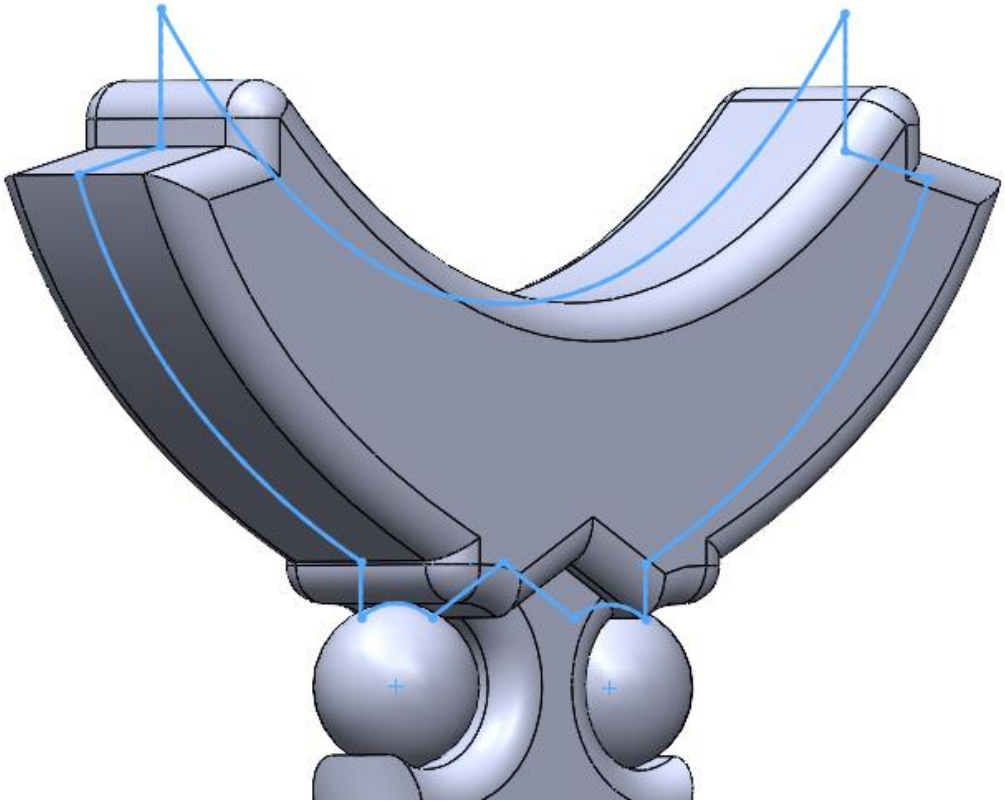


Figure 1.4 - Guard

At the center of Guard creates the rhombus. Firstly, sketch the sides of figure. Then draws the point on another plane, at a distance 1.5 mm. In order to make a figure three-dimensional, you must merge the edges of the figure with a point using the loft boss/base feature.

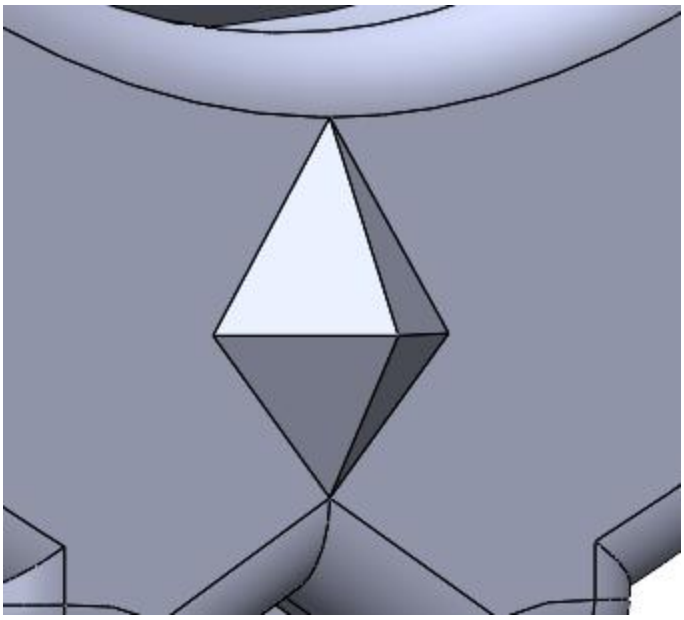


Figure 1.5 - Rhombus

The grip section I divided to two parts. The first one I created by using revolve boss/base feature on two-dimensional sketch. The high of first part is 6.28mm, and the radius of the top is 3mm.

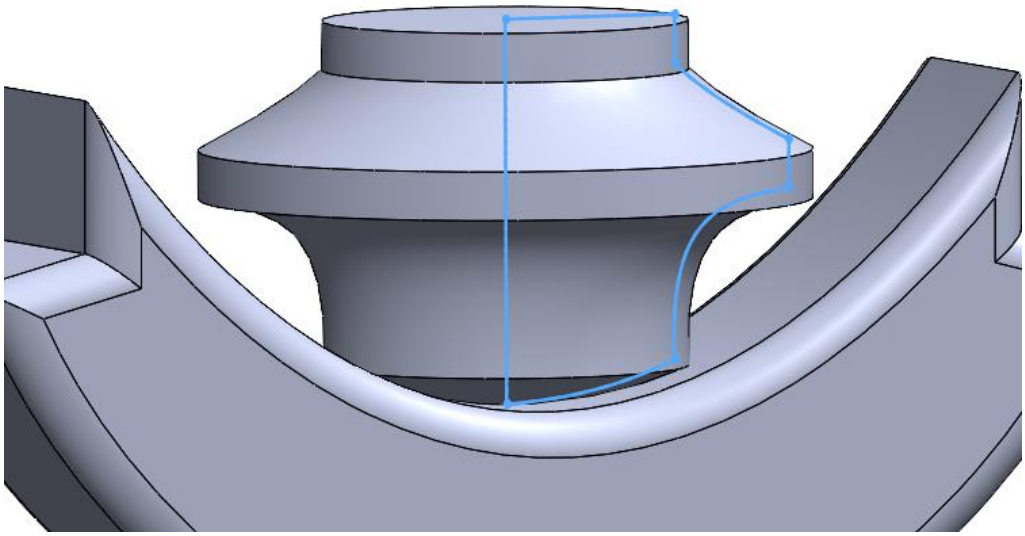


Figure 1.6 – Grip

The second part of the handle is circle, which pulled up to 17 mm. For more detailed handle, I used spiral with swept cut. The last part of full hilt is pommel. The knob on the hilt, which created by using revolve boss base feature.

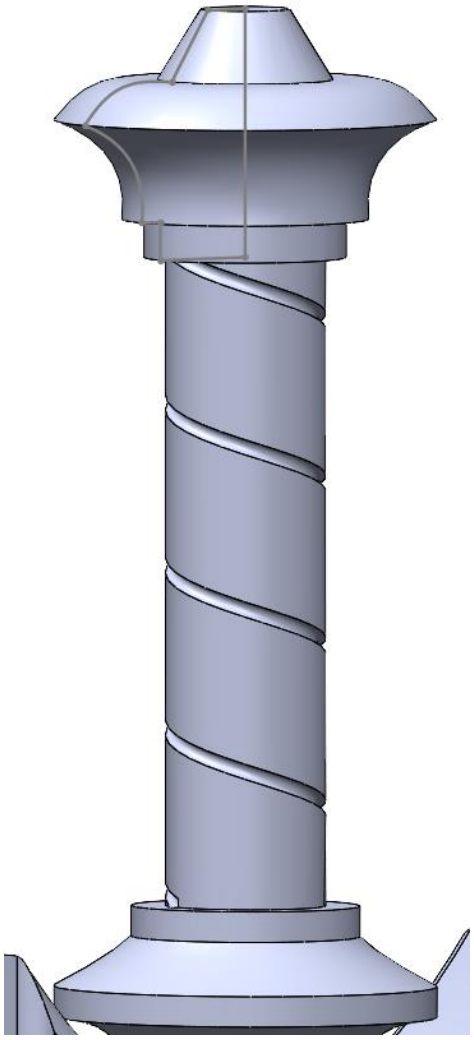


Figure 1.7 – The handle

The final result came out to 110mm in length.

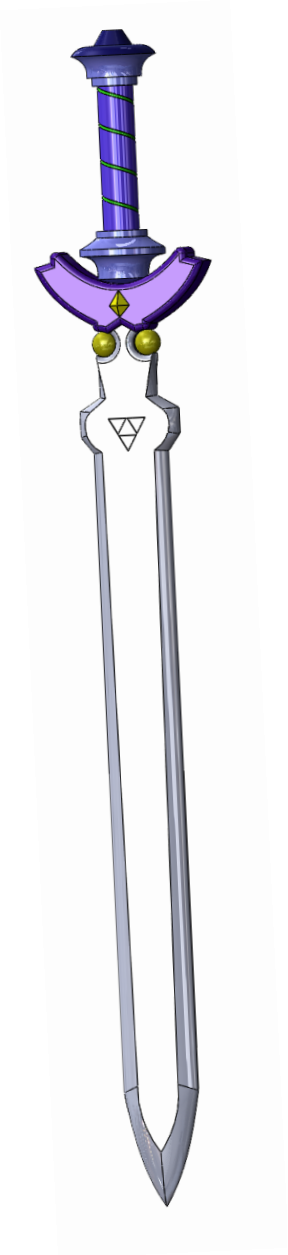


Figure 1.8 – Final Result

Conclusion

This project contains all the features that were covered during the semester. The sword is a fairly simple figure compared to firearms. However, creating this weapon took me much longer than expected, about 10 days at 3 hours a day, considering that the initial sketch was done by hand on paper.